

Theorem

1. for each CFG G , there exists a PDA P st $L(P) = L(G)$
2. for each PDA P , there exists a CFG G st $L(G) = L(P)$

CFG $\rightarrow$ PDA

PDA $\longrightarrow$ CFG

simple
PDA

Definition

A PDA $P = (K, \Sigma, \Gamma, \Delta, s, \bar{F})$ is simple

- (1) $|\bar{F}| = 1$, and
- (2) for each transition $((p, a, \alpha), (q, \beta)) \in \Delta$,
either $\alpha = e, |\beta| = 1$
or $|\alpha| = 1, \beta = e$

Lemma

Every PDA has an equivalent simple PDA.

Proof:

Let $P = (K, \Sigma, \Gamma, \Delta, s, \bar{F})$ be a PDA.

1. If $|\bar{F}| > 1$,

create a new state f
for $q \in \bar{F}$,

create a new transition $((q, e, e), (f, e))$

$$F = \{f\}$$

- Violation:
- 2.1 $|\alpha| \geq 1$ and $|\beta| \geq 1$ ✓
 - 2.2 $|\alpha| > 1$ and $\beta = e$ ✓
 - 2.3 $\alpha = e$ and $|\beta| > 1$ ✓
 - 2.4 $\alpha = e$ and $\beta = e$ ✓

2. for each $((p, a, \alpha), (q, \beta))$ with $|\alpha| \geq 1$ and $|\beta| \geq 1$,
 create a new state r
 replace $((p, a, \alpha), (q, \beta))$ with $((p, a, \alpha), (r, e))$
 $((r, e, e), (q, \beta))$

3. for each $((p, a, \alpha), (q, e))$ with $\alpha = c_1 c_2 \dots c_k$ for some $k \geq 2$
 create $k-1$ new state $r_1, \dots, r_{k-1}$
 replace $((p, a, \alpha), (q, \beta))$ with $((p, a, c_1), (r_1, e))$
 $((r_1, e, c_2), (r_2, e))$
 $\vdots$
 $((r_{k-1}, e, c_k), (q, e))$

4. for each $((p, a, e), (q, \beta))$ with $\beta = c_1 \dots c_k$ for some $k \geq 2$,
 create $k-1$ new states $r_1, \dots, r_{k-1}$
 replace " " with $((p, a, e), (r_1, c_k))$
 $((r_1, e, e), (r_2, c_{k-1}))$

$((q_1, e, e), (q, e))$

5. Create a new symbol $\$$

for each $((p, a, e), (q, e)) \in \Delta$,

create a new state r

replace " " with $((p, a, e), (r, \$))$
 $((r, e, \$), (q, e))$

PDA $\rightarrow$ simple PDA $\rightarrow$ CFG

Lemma

Every simple PDA has an equivalent CFG.

Proof:

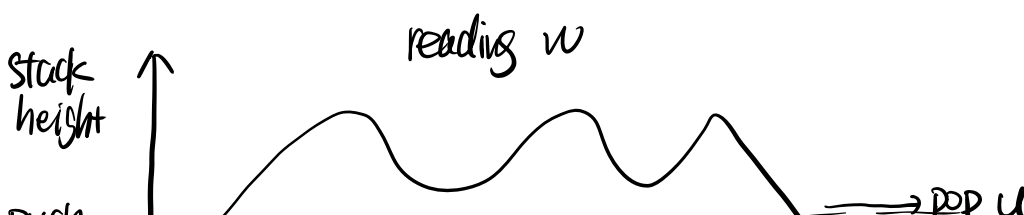
Let $P = (K, \Sigma, T, \Delta, s, f)$ be a simple PDA.

Construct $G = (V, \Sigma, s, R)$ st $L(G) = L(P)$

Non-terminals: $\{ A_{pq} \text{ for } p \in K \text{ and } q \in K \}$

A_{pq} represents $\{ w \in \Sigma^* : (p, w, e) \vdash_P^* (q, e, e) \}$

$S = A_{sf}$ // $L(G) = L(P)$ $\{ w \in \Sigma^* : (s, w, e) \vdash_P^* (f, e, e) \}$

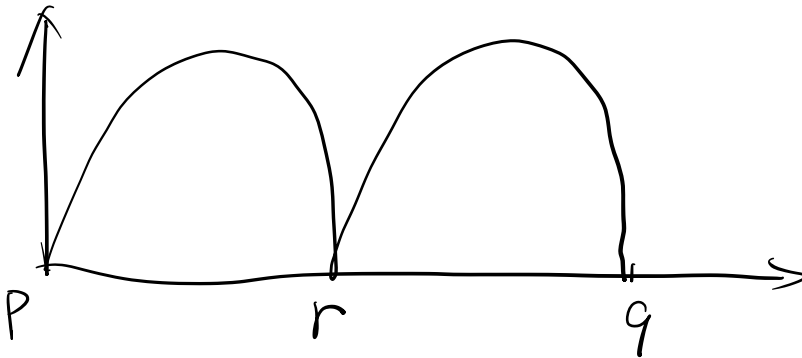
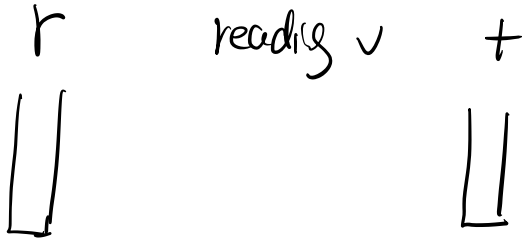




$$w = a v b \text{ for some } v \in \Sigma^*$$

$$(r, v, e) \vdash^* (t, e, e)$$

A_{rt}



$$R = \{ A_p q \rightarrow a A_r t b \text{ for each } (p, a, e), (r, u) \in \Delta \text{ and } (t, b, u), (q, e) \in \Delta \}$$

$$A_p q \rightarrow A_r A_r q \text{ for each } r \in K$$

$$A_p p \rightarrow e \text{ for each } p \in K$$

?

$$S \rightarrow A$$

$$A \rightarrow Aa/e$$

$$L(G) = a^*$$

$$S \rightarrow B$$

$$B \rightarrow BB$$

redundant

claim

$A p q \Rightarrow^* w \in \Sigma^*$ iff $(p, w, e) \vdash^* (q, e, e)$

Proof:

$\Rightarrow$) by induction on # steps of derivation

$\Leftarrow$) " computation

CFG $\rightarrow$ PDA

PDA $\rightarrow$ simple PDA $\rightarrow$ CFG

Corollary

A language is context-free iff it is accepted by some PDA.

Closure property

	$A \cup B$	$A \circ B$	A^*	$A \cap B$	$\bar{A}$
regular	✓	✓	✓	✓	✓
context-free	✓	✓	✓	x	x

A $G_A = (V_A, \Sigma_A, S_A, R_A)$

B $G_B = (V_B, \Sigma_B, S_B, R_B)$

• $A \cup B$ $G_u = (V_u, \Sigma_u, S_u, R_u)$ $L(G_u) = A \cup B$

$S_u \rightarrow S_A \mid S_B$



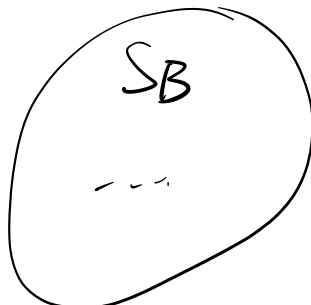
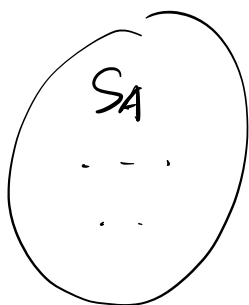
$$V_u = V_A \cup V_B \cup \{S_u\}$$

$$\Sigma_u = \Sigma_A \cup \Sigma_B$$

$$R_u = R_A \cup R_B \cup \{(S_u, S_A), (S_u, S_B)\}$$

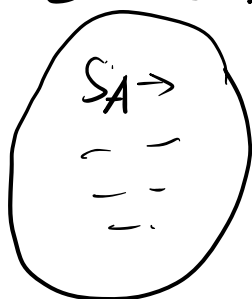
• $A \cdot B$

$$S \rightarrow S_A S_B$$



• A^*

$$\begin{aligned} S &\rightarrow e \\ S &\rightarrow S S_A \end{aligned}$$



• $A \cap B$ not closed

$$A = \{a^i b^j c^k : i=j\}$$

$$B = \{a^i b^j c^k : j=k\}$$

$$A \cap B = \{a^n b^n c^n : n \geq 0\} \text{ — non-context-free}$$

• $\overline{A}$ not closed

$$A \cap B = \overline{\overline{A} \cup \overline{B}}$$

Pumping Theorem (for CFL)

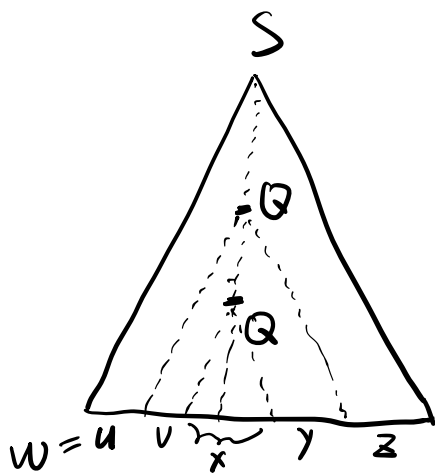
Let L be a CFL. There exists an integer $p > 0$ such that any string in L with length at least p can be divided into five pieces $w = uvxyz$ satisfying the following conditions.

- (1) $uv^i xy^i z \in L$ for any $i \geq 0$
- (2) $|v| + |y| > 0$, and
- (3) $|vxy| \leq p$

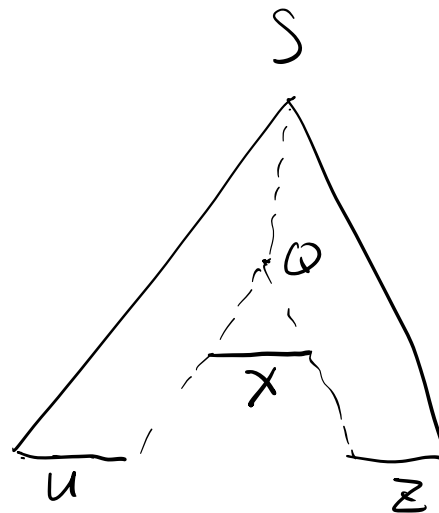
Idea

$$G = (V, \Sigma, S, R)$$

pick a long w from L

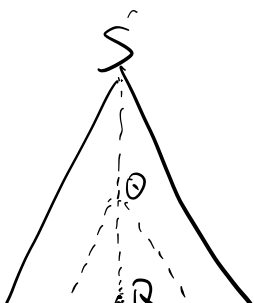


$$\begin{aligned} S &\Rightarrow^* uQz \\ &\Rightarrow^* uvQyz \\ &\Rightarrow^* wxyz \end{aligned}$$

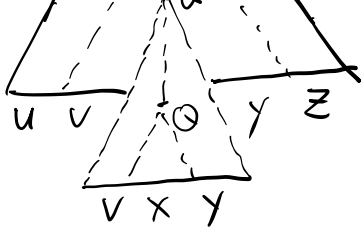


$$\begin{aligned} S &\Rightarrow^* uQz \\ &\Rightarrow^* uxz \end{aligned}$$

$$uv^0xy^0z$$



$$\begin{aligned} S &\Rightarrow^* uQz \\ &\Rightarrow uvQyz \end{aligned}$$



$$\Rightarrow uv^2xy^2z$$

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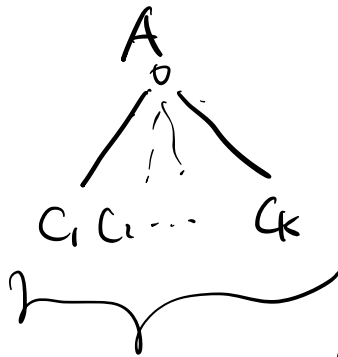
$$uv^2xy^2z \in L$$

$$uv^2xy^2z \in L$$

Proof:

Let $G = (V, \Sigma, R, S)$ be a CFG generating L .

Let $b = \max \{ |u| : (A, u) \in R \}$



$$A \rightarrow C_1 C_2 \dots C_k$$

at most b children

$\Downarrow$

fanoout of any parse tree $\leq b$

o

Fact: If a tree of height h has fanoout $\leq b$,

$$\# \text{ leaves} \leq b^h$$

$\Downarrow$

for a string $w \in L$, the height of its parse tree

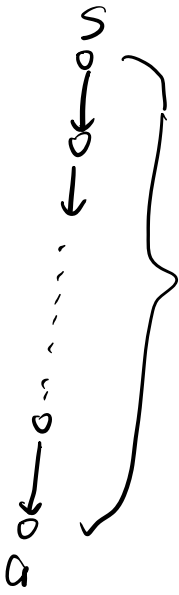
$$\text{is} \geq \log_b |w|$$

$$p := b^{|V-\Sigma|+1}$$

$w :=$ any string in L with $|w| \geq p$

$T :=$ a parse tree of w (with smallest number of nodes)
 height of $T \geq \log_b |w| \geq \log_b b^{|V-Z|+1} = |V-Z|+1$

longest
path

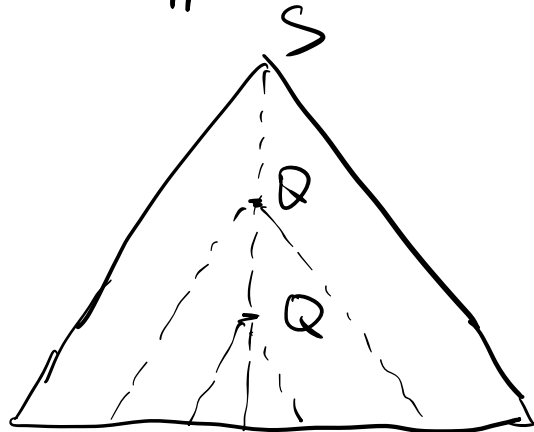


$$\# \text{ edge} \geq |V-Z|+1$$

$$\# \text{ node} \geq |V-Z|+2$$

$$\# \text{ non-terminals} \geq |V-Z|+1 \quad G \text{ has } |V-Z| \text{ distinct non-terminals}$$

$\Downarrow$
 Some non-terminal Q
 appear at least twice.

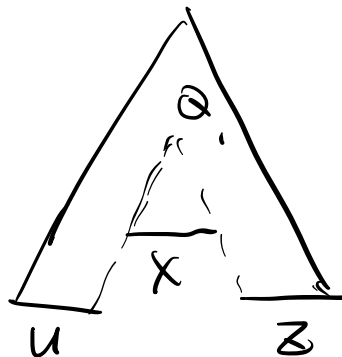


$$w = u v x x z$$

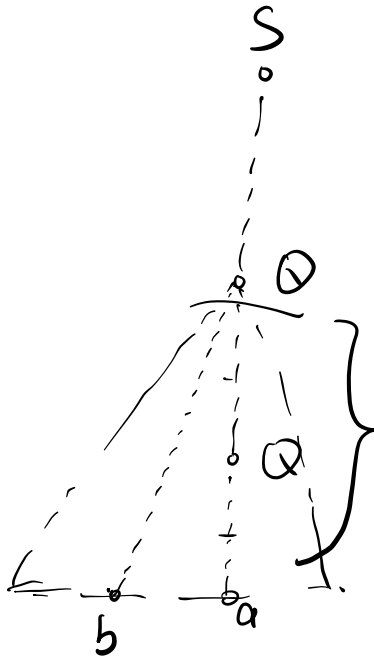
$$(1) u v^i x y^i z \in L \text{ for any } i \geq 0$$

✓

$$(2) |v| + |y| > 0. \text{ If } v = y = e, w = u v x y z = u x z$$



$$(3) |V_{xy}| \leq p$$



every non-terminal appears at most once



subpath from Q to a has at most $|V_{\Sigma}|+1$ edges

the subpath is the long path in the subtree rooted at Q
(from Q to a leaf.)



subtree rooted at Q

has height $\leq |V_{\Sigma}|+1$



leaves of the subtree is $b^{|V_{\Sigma}|+1} = p$
 $|V_{xy}| \leq p$